

Study in the ecosystem structure and trophodynamics in the Kuroshio-Oyashio Extension area

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ARTICLE INFO

Keywords:

Chub mackerel
Ecological network analysis
Ecopath
Ecosystem model
The Kuroshio-Oyashio Extension
The Northwest Pacific

ABSTRACT

The Northwest Pacific (NWP) is not only the region with the highest fishery production in the world, but also the location of the Kuroshio-Oyashio Extension (KOE). Regular monitoring and assessment of marine ecosystem are essential for maintaining its stability and the sustainability of fishery products. Based on the survey data from the high-seas fishery resources survey in the KOE in 2023, the trophic structure and characteristics of open ocean ecosystem in this area were analyzed using the Ecopath with Ecosim model. A total of 25 functional groups were defined based on their ecological characteristics. The field sampling information was used as the biomass data to maximize the quality of the model. The results indicated that the trophic levels of these groups ranged from 1.00 to 4.35. Chub mackerel is a major economic species in the Northwest Pacific Ocean, and the ecotrophic efficiency of chub mackerel was lower than in previous studies (0.74 in 2019), indicating that the proportion of its production contributing to the ecosystem energy might have decreased. Mixed trophic impact results showed that the main interactions between functional groups are primarily predation and feeding competition. Total primary production/total respiration (TPP/TR), and total primary production/total biomass (TPP/TB) indicated the system maturity. Compared to previous studies in the KOE and its adjacent area ecosystems (the total system throughput was 3529.54 t·km⁻²·year⁻¹ and the TR was 486.42 t·km⁻²·year⁻¹), both the TST (1766.22 t·km⁻²·year⁻¹) and the TR (151.27 t·km⁻²·year⁻¹) were relatively lower, which revealed the low productivity characteristic of the KOE ecosystem. Combined with the high TPP/TR (7.31) and the low system omnivory index (0.20), these overall ecosystem characteristics suggested that the KOE ecosystem was structurally simple and vulnerable to disturbance from external activities. Sensitivity analysis indicated that the main output parameters of the model were highly sensitive to the changes in the biomass of the functional groups. This study provided a reference for ecosystem-based fisheries management in the open ocean waters of the KOE.

1. Introduction

The Northwest Pacific (NWP), situated in the East Asian monsoon region, is significantly influenced by the monsoon and features a complex ocean current system (Fig. 1). The North Equatorial Current (NEC) flows westward, while its eastern offshoot, the Kuroshio Current (KC), and the Oyashio Current (OC) to the north converge with the KC to form the Kuroshio-Oyashio Extension (KOE) (Tomita and Kubota, 2005; Ji, 2013). The nutrients carried by the monsoon currents provide abundant nourishment, supporting a highly productive ecosystem. As a result, the Northwest Pacific (NWP) is one of the most productive fishing zones

identified by the International Food and Agriculture Organization (FAO), accounting for approximately a quarter of global fish production (FAO, 2023). Landings in the NWP have been declining annually since peaking in the 1990s, with a continuous downward trend in recent years (FAO, 2022). How to more effectively protect fishery resources in the NWP has become a major concern of researchers and managers. According to the ecosystem-based fisheries management (EBFM), fluctuations of fishery resources can impact other organisms in the ecosystem to varying degrees through complex trophic interactions. Declines in landings of fishery target species may cause fluctuations in stocks of major forage fish species such as mackerel and sardines in the NWP and

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<https://doi.org/10.1016/j.ecolmodel.2025.111152>

Received 4 September 2024; Received in revised form 9 April 2025; Accepted 16 April 2025

Available online 21 April 2025

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consequently affect the stability of the ecosystem. Therefore, the structure and characteristics of the NWP ecosystem need to be continuously monitored to better understand its dynamics. This monitoring could provide a baseline reference for local fisheries management organizations to develop precautionary measures for conservation and management, ultimately offering technical support and reference data for EBFM.

Ecosystem models, such as Ecopath with Ecosim (EwE), are key tools in the implementation of EBFM. These models are based upon the thermodynamic, mass-balanced trophic exchange between living groups and thus offer a robust, quantifiable platform for marine ecosystem research and assessment (Mace, 2001). Compared to single-species models, ecosystem models can more realistically reflect the dynamics of marine ecosystem. These models account for a range of ecological processes and mechanisms, such as trophic interactions among species, changes in environmental conditions in the habitat, and ecosystem structure (Plagányi, 2007). The EwE model is not only user-friendly and requires minimal data, which has led to its extensive application in marine ecosystem studies, but it also offers valuable insights for interdisciplinary research areas such as water resource management and climate change. This model has proven-to-work routines and practices that can be effectively scaled up and adapted to various contexts (Jakeman et al., 2024). For example, the extension of Ecopath in the R language, i.e., Rpath, has enhanced the model's functionality and applicability (Lucey et al., 2020; Wang et al., 2024). With the development of EwE, hundreds of marine ecosystem models have now been established worldwide. The number of ecosystem models for the NWP has also gradually increased (e.g., Murase et al., 2016; Watari et al., 2019; Booth et al., 2020). However, most of these studies have focused on the Pacific coast, while the structure and function of open ocean ecosystems in the NWP, particularly in the KOE, remain insufficiently studied.

The accuracy of model parameters is a primary measure of the quality of ecosystem models (Ainsworth and Walters, 2015). However, due to the scarcity of ocean survey data, biomass parameters in ecosystem models are often difficult to obtain through direct sampling. Instead, these parameters were frequently estimated using data from other models or stock assessment studies, reports from fisheries management organizations, or even the Ecopath model itself (Heymans et al., 2016; Murase et al., 2016).

Based on investigations conducted by the research vessel Songhang from Shanghai Ocean University, we have a series of biomass data that can be used to replace reference or estimated data in the model construction process. We hope this data will enhance the representativeness of the ecosystem model and minimize the uncertainty as much as possible. This study focuses on the open ocean ecosystem of KOE, which is a major area of fishery disturbance in the NWP. The Ecopath model of the KOE was developed using trawl data from the comprehensive investigation conducted by the Songhang. The objectives of this study are: (1) compare with historical findings to assess the change of KOE ecosystem structure and the degree of vulnerability to disturbance, (2) analyze the impact of uncertainty in the main model input parameters on model-derived indices, and (3) to provide a reference for further EBFM in this region.

2. Materials and methods

2.1. Sampling area and data sources

The study area was located in the NWP, and close to the exclusive economic zone (EEZ) of Japan. Range of the study area were 148°E to 164°E and 35°N to 45°N (Fig. 1). A total of 76 fixed survey stations were established in this area. The investigation was conducted using the research vessel “Songhang”, with a gross tonnage of approximately 3000 t. The survey was conducted from June to August 2023, covering a total of 46 days. Biological samples were collected using a fine-mesh pelagic trawl equipped with a four-panel net. The dimensions of the trawl were 434 × 97.1 m, consisting of a 40 mm stretch cod-end mesh and a single capsule structure. Due to workload limitations of the vessel, 36 uniformly distributed “investigative stations” from the total 76 designed trawl survey stations were sampled. In each trawl survey station, the sampling duration was approximately 1–2 h, with a trawling speed of approximately 3–4 knots. A trawl sounder was applied to monitor information on trawl deployment position, trawl mouth opening size, as well as other relevant details during trawling. The trawl depths range from 50 m to 150 m. All catches (mainly fish and cephalopods) were classified and counted, with the total weight of each species was measured on-site.

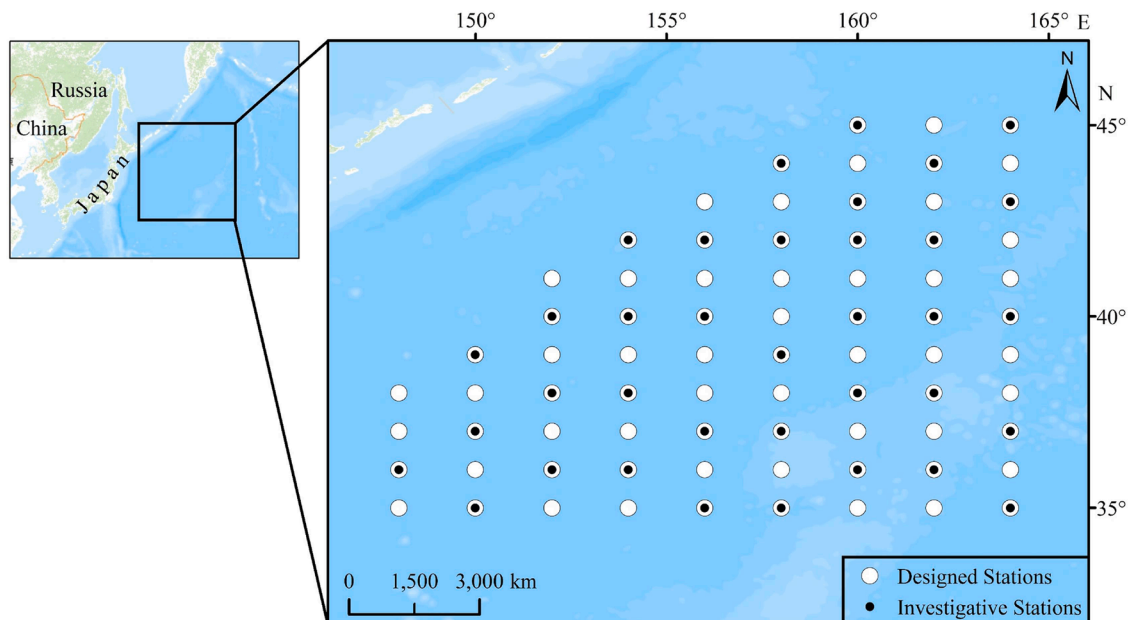


Fig. 1. Sampling station locations of fishery trawls in the Kuroshio-Oyashio Extension.

2.2. Ecopath model

The Ecopath model defines an ecosystem as composed of a set of functional groups, each consisting of species with similar ecological characteristics. It follows thermodynamic principles to study the ecosystem energy flow and network analysis (Plagányi, 2007). Functional groups may include species with the same ecological habits, or they may include species that are economically or ecologically similar. Functional groups may also be used to represent different size or age groups of these species. The full set of functional groups should cover all major energy flow pathways through the ecosystem (Christensen and Pauly, 1992; Heymans et al., 2016). Ecopath estimates the linkage of trophic mass-balance using biomass (B), production/biomass (P/B), consumption/biomass (Q/B), ecotrophic efficiency (EE), and diet composition data as input parameters (Christensen et al., 2005; Heymans et al., 2016; Lucey et al., 2020; Taghavimotlagh et al., 2021). The basic equation for model construction was expressed as:

$$B_i \times \left(\frac{P}{B}\right)_i \times EE_i = \sum_{j=1}^n B_j \times \left(\frac{Q}{B}\right)_i \times DC_{ji} + Y_i + BA_i + E_i$$

Where B_i and B_j is the biomass of functional group i and j ; $(P/B)_i$ is production rate, which can generally be replaced by the total mortality of functional group i ; $(Q/B)_i$ is consumption rate of functional group i ; EE_i is ecotrophic efficiency of functional group i ; DC_{ji} is proportion of i in the diet of j ; Y_i is catch; BA_i is biomass accumulation rate for functional group i ; E_i is the net migration rate.

2.3. Functional groups

The aquatic organisms in the KOE ecosystem were classified into 25 functional groups: 1 detritus group (particulate organic matter, POM), 1 primary producer group (phytoplankton), 1 zooplankton group, 18 fish groups, 2 cephalopoda groups, and 2 crustacean micronekton groups. The composition of each group is listed in Table 1.

2.4. Basic input parameters quality

Biomass estimates for functional groups were derived from the 2023 Songhang survey in 2023, literature review, and other relevant studies in the KOE and adjacent areas. The swept area method was used to estimate the biomass of most fish species using data from the trawl survey in the KOE area, 2023. The swept area is obtained by multiplying the effective width of the fishing gear by the distance traveled by the fishing vessel during the operation. The biomass per unit area is then calculated as the ratio of the total catch to the swept area. However, due to limitations in gear selectivity, some fish, such as tunas, could not be effectively sampled. Consequently, the biomass of tunas was estimated based on the latest report released by International Scientific Committee (ISC) (ISC, 2023). The biomass of phytoplankton was estimated from survey data, while the biomass of zooplankton and detritus was referenced to relevant studies. Specific data and parameter sources for the model were detailed in the Supplementary Table S1.

Consumption rates, Q/B, were derived from an empirical function based on caudal fin profile (Palomares and Pauly, 1989) and obtained from FishBase (<https://www.fishbase.org>) and from other relevant studies. Production rates, P/B, were obtained from relevant studies or estimated based on the Production/consumption (P/Q) proposed by Christensen et al. (2005). Diet composition data were obtained from recent studies analyzing stomach contents in the same or adjacent areas. Quantitative diet data in terms of wet weight was used for most functional groups, while diets for some groups were described qualitatively. The fraction of functional group production used within the living food web, ecotrophic efficiencies (EE), were estimated by the model. Due to gear selectivity limitations, accurately estimating the biomass of some smaller sized organisms were challenging. The biomass of these groups

Table 1

Species composition of each functional groups for the Kuroshio-Oyashio Extension ecosystem model.

Group name	Main species composition
Blue shark	<i>Prionace glauca</i>
Other sharks	<i>Isurus oxyrinchus</i> , <i>Lamna ditropis</i>
Swordfish	<i>Xiphias gladius</i>
Tunas	<i>Thunnus alalunga</i>
Pelagic stingray	<i>Pteroplatytrygon violacea</i>
Ocean sunfish	<i>Mola mola</i>
Salmon	<i>Oncorhynchus kawamurae</i> , <i>O. keta</i> , <i>O. gorbuscha</i>
Large mesopelagic fishes	<i>Alepisaurus ferox</i> , <i>Ichtyosaurus aenigmaticus</i> , <i>Nemichthys scolopaceus</i> ,
Large pelagic fishes	<i>Desmodema polystictum</i> , <i>Diplospinus multistriatus</i> , <i>Gempylus serpens</i> , <i>Lampris guttatus</i> , <i>Tradhypterus ishikawae</i>
Small mesopelagic fishes	<i>Astronesthes</i> spp., <i>Diplophos taenia</i> , <i>Dolicholagus longirostris</i> , <i>Howella parini</i> , <i>Lestidiops jayakari</i> , <i>Lestrolepis intermedia</i> , <i>Lipolagus ochotensis</i> , <i>Nealotus tripes</i> , <i>Paracaristius maderensis</i> , <i>Scopelosaurus hoedti</i> , <i>Stomias affinis</i>
Small pelagic fishes	<i>Lactoria cornutus</i> , <i>Leptocephalus</i> , <i>Trachurus japonicus</i>
Pomfret	<i>Brama brama</i> , <i>Brama myersi</i> , <i>Brama japonica</i>
Medusafish, barrelfish and driftfish	<i>Cubiceps</i> spp., <i>Hyperoglyphe japonica</i> , <i>Ichthyos lockingtoni</i> , <i>Psenes arafurensis</i> , <i>P. pellucidus</i> , <i>Tetragonurus</i> spp.
Chub mackerel	<i>Scomber australasicus</i> , <i>S. japonicus</i>
Japanese anchovy	<i>Engraulis japonicus</i>
Pacific sardine	<i>Sardinops melanostictus</i>
Pacific saury	<i>Cololabis saira</i>
Lantern fish	<i>Ceratoscopelus townsendi</i> , <i>Diaphus gigas</i> , <i>D. perspicillatus</i> , <i>Lampanyctus</i> spp., <i>Myctophum asperum</i> , <i>M. nitidulum</i> , <i>Notoscopelus caudispinosus</i> , <i>N. japonicus</i> , <i>N. resplendens</i> , <i>Symbolophorus californiensis</i> , <i>Tarletonbeania taylori</i>
Cephalopoda	<i>Enoplateuthis</i> spp., <i>Eucleoteuthis luminosa</i> , <i>Gonatopsis octopodatus</i> , <i>Leachia</i> spp., <i>Moroteuthis robusta</i> , <i>Octopoteuthis sicula</i> , <i>Ommastrephes bartrami</i> , <i>Onychoteuthis compacta</i> , <i>Ornithoteuthis volatilis</i> , <i>Taonius pavo</i> , <i>Todarodes pacificus</i>
Small cephalopoda	<i>Cranchia scabra</i> , <i>Japetella diaphana</i> , <i>Watasenia scintillans</i>
Krill	<i>Euphausia recurva</i> , <i>E. nana</i> , <i>E. pacifica</i> , <i>E. mutica</i> , <i>E. similis</i>
Other crustaceans	Amphipoda
Zooplankton	Copepoda, Ostracoda
Phytoplankton	<i>Thalassiosira</i> spp., <i>Navicula</i> spp., <i>Dinophysis</i> spp.
Detritus	particulate organic matter (POM)

was estimated by Ecopath model (Christensen et al., 2005) assuming that EE for these groups (ocean sunfish, small mesopelagic fish, small pelagic fish, anchovy, Pacific saury, cephalopoda and small cephalopoda) was 0.95. Specific data and parameter sources for the model were detailed in the Supplemental material (Supplementary Table S1, Table S2).

2.5. General ecosystem indices

Mixed trophic impact analysis (MTI) was utilized to evaluate the net positive or negative between all functional groups within a stable ecosystem (Ulanowicz and Puccia, 1990; Christensen et al., 2005). MTI was calculated by constructing an $n \times n$ matrix, where i and j represent the interactions between impacted group i and impacting group j :

$$MTI_{ij} = DC_{ij} - FC_{ij}$$

where DC_{ij} is the diet composition term which expresses how much species i contributes to the diet of j , while FC_{ij} represents the feeding proportion of i as a predator on j .

Additionally, a range of ecological indices were employed to assess the overall condition of the open ocean KOE ecosystem (Christensen et al., 2005; Odum, 1969): (1) Total system throughput (TST, $t \cdot km^{-2} \cdot year^{-1}$), total exports, total respiration, and total biomass indicated the size of an ecosystem; (2) Net system production (NSP), total primary production/total respiration (TPP/TR), and total primary

production/total biomass (TPP/TB) were used for judging the maturity of the system; (3) system omnivory index (SOI) reflected the complexity of the food web within the ecosystem.

2.6. Pre-balancing and balancing analyses

PREBAL provides a set of diagnostic tools based on ecological and fisheries principles to check the reasonableness of input data and ensure that key parameters are ecologically viable before the model is balanced, and it can provide an important basis for model balancing and subsequent analysis (Link, 2010; Heymans et al., 2016). For example, PREBAL checks production to biomass (P/B) ratios to ensure that they are ecologically appropriate (the P/B ratios of predators are typically lower than those of their prey). In accordance with the second law of thermodynamics, PREBAL checks the P/Q ratio to ensure that it is between 0.1 and 0.3 (Christensen et al., 2005; Darwall et al., 2010).

The pedigree index provides an error range on parameters which can be used in Monte Carlo resampling based on a semi-quantitative ordinal score of data quality (Funtowicz and Ravetz, 1990; Christensen et al., 2005). Each source of input parameters for each functional group required allocation of a different 'quality score'. Each score was allocated a specific (and default) Confidence Interval, CI (± 10 –90 % CI). Default CIs were obtained from the pre-defined pedigree tables provided in Christensen et al. (2005) (Supplementary Table S3) and were utilized to generate parameter-specific pedigree indices, or CI, for B, P/B and Q/B (Supplementary Table S4).

2.7. Sensitivity analysis

The Monte Carlo method was applied to analyze the impact of uncertainty in the main model input parameters (biomass, P/B, and Q/B of the functional groups) on model-derived indices (TST, total production (TP), net primary production (NPP), TPP/TR, and TPP/TB). First, different levels of uncertainty for the input parameters (10 %, 25 %, 40 % for biomass; 10 %, 20 %, and 30 % for P/B and Q/B) were set based on the CI of input parameters sources for each functional group (Morissette L., 2007). Then the simulated values were randomly resampled from a mean distribution ranging from [(1-uncertainty level) $\times$ original input value, (1+uncertainty level) $\times$ original input value] each time. The results of corresponding model projections were then estimated and compared. This process was repeated 1000 times. Since changes in input parameters could lead to an unbalanced situation, only resampled values that satisfied the model's balance conditions (based on the $EE < 1$) were selected for subsequent analysis (the range, proportion, and CV of successful test parameters were recorded simultaneously). Then we calculated the range of the model-derived indices through uncertainty analysis. Impacts were assessed as changes (relative bias) to model-derived ecosystem indices:

$$RB = \frac{Y_{\text{monte carlo}} - Y_{\text{base}}}{Y_{\text{base}}} \cdot 100\%$$

Where *RB* is the relative bias for each index, $Y_{\text{monte carlo}}$ is the index value derived from the randomly resampled model parameter set, and Y_{base} is the index value derived from the originally defined parameter set.

In this study, the Ecopath model for the KOE was constructed using Ecopath with Ecosim 6.6.8 software. ArcMap 10.2 was used to plot the survey stations map. The ggplot2 package in R 4.3.2 was used to plot the parameter sensitivity analyses result. While the parameter sensitivity analysis were mainly performed using the Rpath package in the R (Wickham H, 2016).

3. Results

3.1. Food web and trophic level characteristics

Trophic levels of functional groups in the 2023 KOE ecosystem ranged from 1.00 to 4.35 (Table 2). The blue shark had the highest trophic level at 4.35, followed by other sharks, swordfish, and tunas, with trophic levels of 4.34, 4.30, and 4.30, respectively. Among the fish, the lowest trophic level was for ocean sunfish (2.66). In addition to most fish groups, cephalopoda and small cephalopoda were also in the trophic level higher than 3 as secondary consumers. The second trophic level includes ocean sunfish, other crustaceans, krill, and zooplankton (Fig. 2).

The ecotrophic efficiency (EE) of all functional groups was less than 1, with the lowest EE for the detritus (0.01), followed by blue shark (0.05), zooplankton (0.06), salmon (0.11), Pomfrets (0.13), other sharks (0.13), swordfish (0.16) and chub mackerel (0.17). The EE of the remaining functional groups was all greater than 0.3. Omnivory indices for each functional group ranged from 0 to 0.79. Cephalopoda had the highest omnivory indices, followed by small cephalopoda (0.58) and large mesopelagic fishes (0.53), while the omnivory indices of all other functional groups were below 0.5.

3.2. Mixed trophic impact analysis

Mixed trophic analyses revealed the net effects of changes among both direct and indirect relationships on each functional group (Fig. 3). As primary producers in the ecosystem, phytoplankton had positive effects on most of the functional groups, while zooplankton had similar effects. In general, most functional groups exerted pronounced positive effects on their predators and direct negative effects on their prey functional groups (e.g., cephalopoda vs sharks, zooplankton vs phytoplankton). Additionally, some functional groups showed negative effects due to similar food composition (e.g., chub mackerel vs small pelagic fish, Pacific sardines vs Pacific saury). As primary prey items, krill, zooplankton, phytoplankton and POM experienced negative impacts from other functional groups, with small pelagic fish and anchovy having higher impacts on them. Some functional groups had a minor impact on others due to their relatively low biomass, such as pelagic stingray, large pelagic fish and pomfret.

3.3. General ecosystem indices

In Ecopath models, general ecosystem characteristics are described by a series of ecological indices that reflect system size, stability, maturity, and so on. The total system throughput (TST) is the sum of all flows in a system. It is estimated as the sum of four components (total consumption, total export, total respiration and total flows to detritus) of the flows. TST of the 2023 KOE ecosystem was 1766.22 t·km⁻²·year⁻¹, of which the sum of all flows into detritus accounted for 37.99 % (670.92 t·km⁻²·year⁻¹). The sum of all consumption (500.07 t·km⁻²·year⁻¹) was the next highest, accounting for 28.31 % of the total system throughput. Calculated total net primary production (NPP) was 1105.50 t·km⁻²·year⁻¹, which was considerably larger than the sum of all respiratory flows, resulting in the TPP/TR greater than 1. The TPP/TB ratio was 55.00, while the system omnivory index (SOI) was 0.20. Nutrient flow in the KOE ecosystems had a high proportion of contribution from food chains with phytoplankton as the base nutrient source. Phytoplankton contributed 74 % of the energy in the KOE ecosystem, while 26 % came from POM detritus (Table 3).

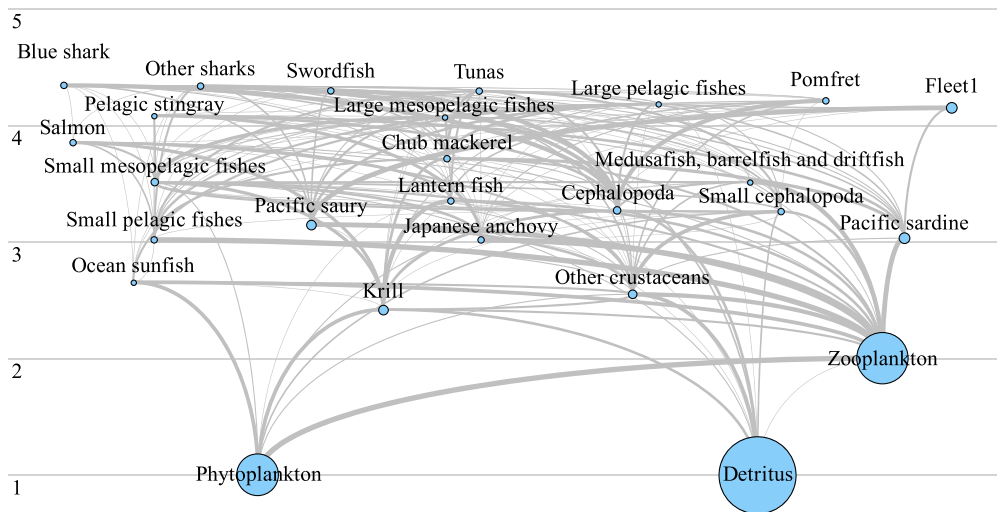
3.4. Model quality evaluation and parameter sensitivity analysis

The Pedigree index of the 2023 KOE ecosystem model was 0.581 (Table 3). Parameter sensitivity analysis results indicated that the value range of major ecosystem indices increased with the uncertainty of the

Table 2

Input parameters of Ecopath model function group for Kuroshio-Oyashio Extension ecosystem.

No.	Group name	Trophic level	Biomass (t/km ²)	Production / biomass (/year)	Consumption / biomass (/year)	Ecotrophic Efficiency	Catch (t/km ²)	Omnivory index
1	Blue shark	4.35	0.0111	0.46	2.75	0.05		0.10
2	Other sharks	4.34	0.0057	0.39	2.50	0.13	0.0003	0.12
3	Swordfish	4.30	0.0117	0.49	2.50	0.16		0.04
4	Tunas	4.30	0.0150	0.42	4.60	0.81	0.0038	0.07
5	Pelagic stingray	4.08	0.0005	0.75	2.50	0.51	0.0002	0.12
6	Ocean sunfish	2.66	0.0022	0.75	2.50	0.95		0.37
7	Salmon	3.85	0.0044	1.94	6.47	0.11		0.23
8	Large mesopelagic fishes	4.07	0.0002	0.45	1.50	0.55		0.53
9	Large pelagic fishes	4.19	0.0008	1.59	5.30	0.60	0.0008	0.17
10	Small mesopelagic fishes	3.51	0.1429	2.50	10.00	0.95	0.0004	0.46
11	Small pelagic fishes	3.02	0.1000	1.52	5.07	0.95	0.0002	0.01
12	Pomfret	4.21	0.0221	0.48	5.80	0.13		0.05
13	Medusafish, barrellfish and driftfish	3.52	0.0006	0.95	4.86	0.79		0.41
14	Chub mackerel	3.72	0.0743	2.79	11.63	0.17	0.0125	0.19
15	Japanese anchovy	3.02	0.1117	2.96	9.87	0.95	0.0001	0.01
16	Pacific sardine	3.03	0.5777	0.90	5.61	0.74	0.1037	0.02
17	Pacific saury	3.15	0.3984	1.25	4.17	0.95	0.4600	0.06
18	Lantern fish	3.35	0.0383	3.20	10.00	0.65		0.04
19	Cephalopoda	3.27	0.1278	2.56	7.30	0.95	0.0006	0.79
20	Small cephalopoda	3.26	0.0525	3.50	13.64	0.95		0.58
21	Krill	2.42	0.4641	2.56	12.05	0.95		0.38
22	Other crustaceans	2.55	0.2702	8.00	30.00	0.95		0.25
23	Zooplankton	2.01	10.4790	23.16	45.35	0.06		0.01
24	Phytoplankton	1.00	7.1890	153.78		0.43		
25	Detritus	1.00	28.5000			0.01		

Note: Bold is the result of model estimation.**Fig. 2.** Food web of the Kuroshio-Oyashio Extension ecosystem.

model input parameters (Fig. 4). However, the variation ranges of the indices were smaller than those of the input parameters. The change in biomass and P/B of functional groups affected all the five model-derived indices (total system throughput, TST; sum of all production, TP; calculated total net primary production, NPP; total primary production/total respiration, TPP/TR; total primary production/total biomass, TPP/TB), but overall, the impact of biomass on the output indicators is greater than that of P/B. Changes in the Q/B ratio of functional groups substantially affected the TPP/TR of ecosystems, while having minimal impact on other indicators. As we increased the parameter space being re-sampled, a larger proportion of the draws result in unbalanced models. Generally, the range of each parameter (B, P/B, and Q/B) closely matched to the original model input parameters range. Increased

confidence intervals (CI) of input parameters were associated with gradual decreases in the proportion of successfully analyzed parameters and rising their coefficient of variation (CV) values. For example, the proportion of successfully analyzed parameters of B ranged from 42.70 % to 100.00 % and the proportion of Q/B ranged from 98.00 % to 100 % (detailed in the **Supplementary Table S5**).

4. Discussion

4.1. Trophic level and transfer efficiency in the KOE ecosystem

For a long time, ecosystem modeling has been a primary tool for assessing the function and stability of ecosystem (Plagányi, 2007).

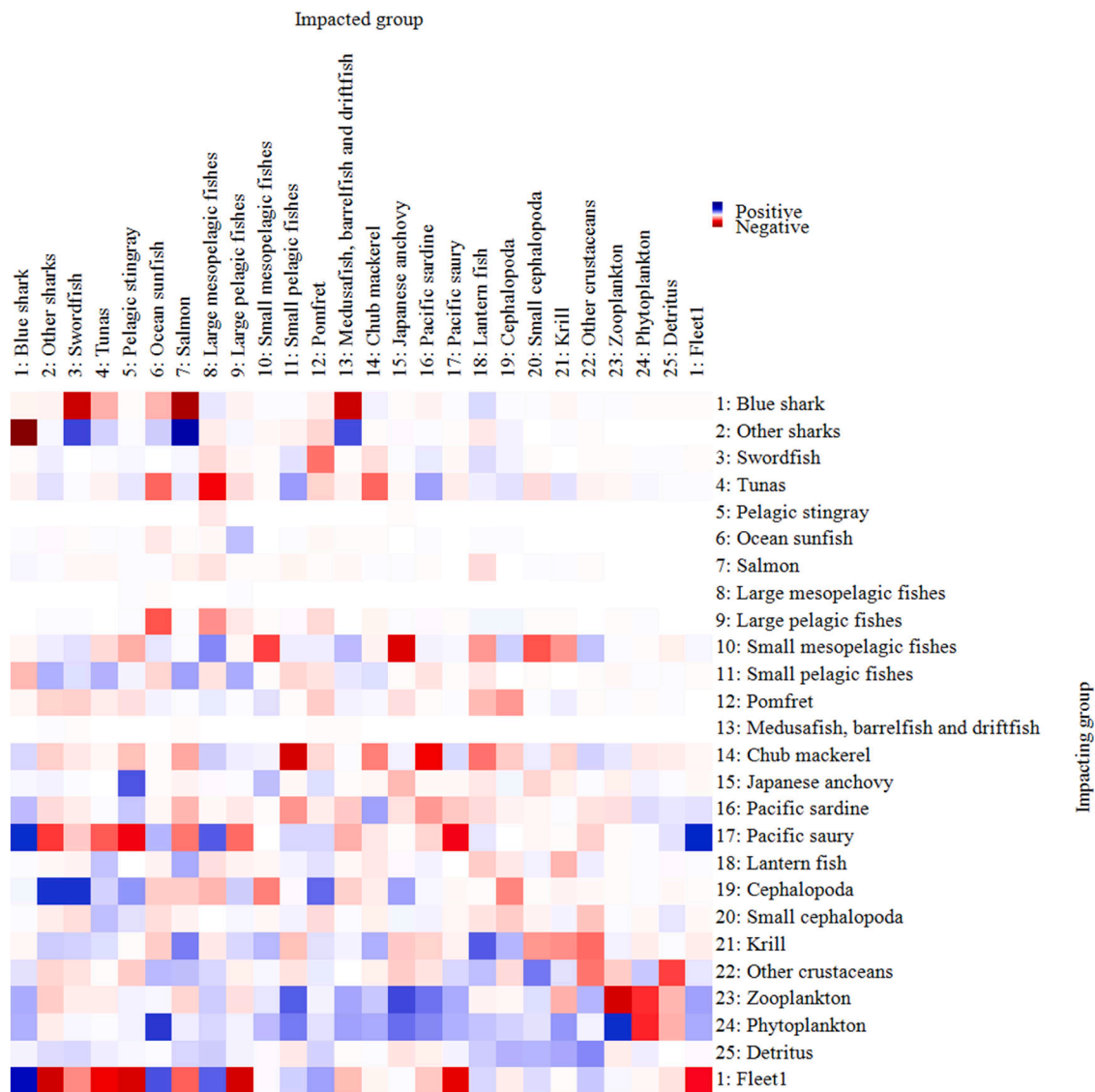


Fig. 3. Mixed trophic impacts between functional groups in the Kuroshio-Oyashio Extension ecosystem.

However, due to the lack of extensive field survey data, ecosystem-level evaluations have been infrequent in the KOE, which is an important fishery area. This limitation undoubtedly hinders the implementation of ecosystem-based fisheries management. Therefore, using biomass survey data from the KOE, we constructed an Ecopath model to assess the trophic structure and energy flow of the KOE ecosystem in 2023. Generally, compared to the trophic structure of other marine ecosystem in the KOE (Murase et al., 2016; Watari et al., 2019), the range of trophic levels of functional groups in this open ocean area was narrower. For example, the trophic levels of most small pelagic fishes, such as chub mackerel, Pacific saury, and Pacific sardine, were higher than those in other studies, while trophic levels of some top predators (sharks and tunas) were lower than other studies. Furthermore, variations in functional groups, diet, and parameters (P/B, Q/B) may also be responsible for these findings, which warrant further thorough investigation to elucidate their underlying mechanisms.

Ecotrophic Efficiency (EE) is commonly used to indicate the utilization efficiency of functional groups (Heymans et al., 2016). EE for most small fishes and cephalopoda was relatively high, similar to findings in other KOE studies (Murase et al., 2016; Watari et al., 2019). However, the EE of the main fishery target species (0.17), chub mackerel, was lower than previous studies in Watari et al. (2019) (0.74 in

2019) but similar to the result of Murase et al. (2016). This indicates that the production of chub mackerel contributes a lower proportion of energy and consumption to the KOE ecosystem. And this result may be caused by the higher biomass of the chub mackerel compared to the other fish functional groups in this model. But it should be note that the biomass of chub mackerel in this study was lower than that of previous studies (Murase et al., 2016; Watari et al., 2019). The low EE of chub mackerel may also be due to unmodelled mortality, which is commonly considered to be natural mortality in addition to predation mortality and fishing mortality (Christensen et al., 2005; Heymans et al., 2016). Meanwhile, since the chub mackerel was a migratory species, the low EE value was also potentially due to partial fishing mortality and predation mortality occurring outside the study area. The above reasons derive from the fact that mortality caused by predation and fishing is relatively weak compared to its own natural mortality. The utilization of chub mackerel in the KOE ecosystem could be sub-divided as 36.09 % from fishery catch and 63.91 % from predation. As a mid-trophic level fish with high abundance, chub mackerel not only has high economic value but also serves as prey for many top predators, which explains why its EE is dominated by predation. Compared with the results from other Ecopath model, the fishing mortality of chub mackerel was lower in this study (Watari et al., 2019). Even though chub mackerel in the KOE

Table 3

General ecosystem indices parameters of Kuroshio-Oyashio Extension ecosystem.

Parameter	Value	Unit
Sum of all consumption	500.07	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
Sum of all exports	443.95	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
Sum of all respiratory flows (TR)	151.27	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
Sum of all flows into detritus	670.92	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
Total system throughput (TST)	1766.22	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
Sum of all production	1354.29	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
Mean trophic level of the catch	3.15	
Calculated total net primary production (NPP)	1105.50	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
Total primary production/total respiration (TPP/TR)	7.31	
Net system production	954.23	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
Total primary production/total biomass (TPP/TB)	55.00	
Total biomass (excluding detritus) (TB)	20.10	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
Total catch	0.58	$\text{t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$
System Omnivory Index (SOI)	0.20	
Ecopath pedigree	0.58	
Proportion of total flow originating from phytoplankton	0.74	
Proportion of total flow originating from detritus	0.26	

currently exhibits high biomass and low EE, its sustainable fishing still warrant attention in fisheries management.

Additionally, some higher trophic level carnivores, such as sharks and swordfish, exhibited lower EE as they have few predators. In most ecosystem models, the EE of sharks is typically low and primarily consisted of fisheries catch (Howell et al., 2013; Booth et al., 2020). Sharks tend to not be primary target species in most fisheries, and their fishing mortality is generally low in most ecosystem (Kindong et al., 2022). As a top predator, there is minimal predation on sharks. Therefore, the low EE of sharks is primarily due to their low predation and fishing mortalities.

4.2. Mixed trophic impact analysis

Mixed trophic impact (MTI) analysis was employed to examine the direct and indirect interactions among functional groups within the ecosystem food webs (Taghavi-motlagh et al., 2021). The MTI results indicated that the main interactions between functional groups are primarily predation and competition. Most functional groups had considerable negative impacts on themselves due to intraspecific competition arising from similar food composition and resource competition. Phytoplankton, zooplankton, and krill, as common basic food organisms, had considerable positive impacts on most functional groups. These prey organisms, dominated by phytoplankton as the primary energy source, drive a considerable portion of the energy flow in the ecosystem. After organisms are transformed into detritus in open ocean areas, only a small portion remains in the ecosystem as POM. Production and most trophic pathways in the KOE ecosystem appear to be largely driven by food chains with phytoplankton as the base energy source.

Most interactions among fish at the mid-trophic level were negative. These interactions might arise from feeding competition among fish at similar trophic levels (e.g., the impact of small mesopelagic fish on anchovy and the impact of chub mackerel on small pelagic fish), or from predation (e.g., the impact of chub mackerel on anchovy). Considerable negative impacts of fishing on higher trophic level fish (e.g., sharks, tunas, pelagic stingray, large pelagic fish) were evident. According to the top-down control theory, top predators are selectively removed from the water by fishing (Song et al., 2007).

A limitation of this study is that it does not account for the role of marine microbes in the chemical cycling of nutrients in ecosystems. Microbes play a crucial role in regulating various bio-element transformations and bioremediation processes (Mcquatters-Gollop et al., 2024). This study primarily focused on high trophic level ecosystems and did not incorporate microbial function groups. In the future, we will include a broader range of functional groups to enhance the completeness of ecosystem model construction.

4.3. General ecosystem indices

Overall, the KOE ecosystem was characterized by a low productivity,

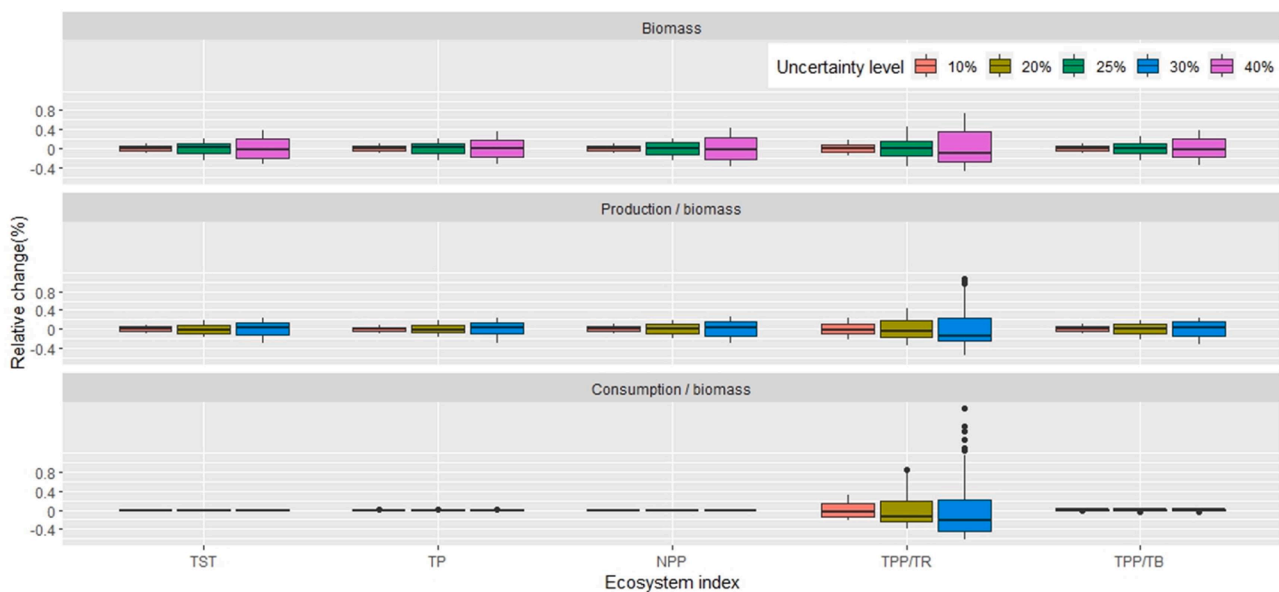


Fig. 4. Parameter sensitivity analysis in the Ecopath model for the Kuroshio-Oyashio Extension. Boxes depict 25th and 75th percentiles, indicating the distribution of the middle 50 % of the data. The horizontal line inside the box indicates the median of the data. The "whiskers" outside the box represent the range of the data, typically from the minimum to the maximum value, but excluding outliers. Data points that lie outside the "whiskers" are considered outliers or extreme values.

low stability, and a relatively simple food web. The Total system throughput (TST) in the KOE ecosystem was $1766.22 \text{ t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$, which was lower in scale compared with the TST reported in the 2019 KOE study ($3529.54 \text{ t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$) (Watari et al., 2019). This may suggest that the KOE ecosystem scale was lower than previous years, but also reflects the fact that the two KOE models cover somewhat different study areas. Our survey stations were located in oceanic waters far offshore, characterized by a relatively simple ecosystem structure and function, and less impacted by land-based anthropogenic activities. Ecosystem maturity reflects the complexity and stability of an ecosystem in terms of its structure and function. Mature ecosystems typically have higher species diversity, complex trophic relationships, efficient energy utilization, and strong resilience (Christensen et al., 2005; Odum, 1969). Net system production (NSP) is the difference between total net primary production (TPP) and sum of all respiratory flows (TR), which indicates a higher ecosystem maturity as the value approaches 0 (Christensen et al., 2005). The Total primary production/total respiration ratio (TPP/TR) is another key indicator of ecosystem maturity. The closer the value of TPP/TR is to 1, the more mature the ecosystem is, where all production is utilized, leaving no excess available for reuse (Odum, 1969). The TPP/TR of the KOE ecosystem model in 2023 was 7.31, with TR at only $151.27 \text{ t}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$, which is considerably lower than reported in most previous studies of the KOE and its adjacent area ecosystems (Jiang et al., 2010; Lin et al., 2013; Watari et al., 2019). This could be attributed to the lower biomass of secondary consumers. Those indicators also indicated that this ecosystem is currently disturbed to a large extent and the stability of the system is still weak. Given that the KOE is an oceanic ecosystem in open ocean area, its increase of variability primarily arises from fishing and environmental variability (Sivel et al., 2022).

The omnivory index (OI) of functional groups is calculated based on the variance in the trophic levels of their prey. Functional groups that feed on a single trophic level will have an OI approaches 0. As the OI approaches 1, the trophic ranges of prey are broader, and the complexity of the food web increases (Christensen et al., 2005). The low OI of most species in the KOE indicated that these species have a specialized diet, feeding mainly on specific species or a few functional groups. For example, salmon primarily fed on small fishes and crustaceans (Sakai et al., 2012). In the KOE ecosystem, functional groups with lower OI are predominantly found among both higher and lower trophic level fishes. For example, top predators such as sharks, tunas, and swordfish had a relatively high concentration of prey organism species (Kubodera et al., 2007; Trujillo-Olvera et al., 2018) thus showing lower OI. Some small pelagic fishes, such as anchovy, Pacific sardine and Pacific saury, primarily feed on zooplankton, resulting in lower OI. In contrast, cephalopoda exhibited a higher OI. This is likely due to their diverse diet. Watanabe et al. (2004) demonstrated that the prey items of cephalopoda varied widely, ranging from zooplankton to large pelagic fish. Such highly generalized diet might also be the main reasons why it can be widespread in the ecosystem, had high capacity to adapt to environmental changes, and became a major fishery species in the KOE (Katugin et al., 2014).

The specialized diet of most species in this study area resulted in a low complexity of the ecosystem and a simpler food web structure. Consequently, the system omnivory index (SOI) of the ecosystem was relatively low (0.20), similar to that of the offshore ecosystem in the KOE (Jiang et al., 2010; Lin et al., 2013). However, it should be noted that the low SOI in most offshore ecosystems may be partly caused by ecosystem degradation due to fishing pressure and environmental change (Heymans et al., 2016), while the SOI in open ocean ecosystems was probably due to the specialized diet of pelagic species.

4.4. Evaluation of overall model uncertainty

Based on the results of PREBAL, the model is considered reasonable, although some parameters require more complete updating in

subsequent studies and further testing of the model. In a survey of 50 mass-balanced food web models, Morissette et al. (2007) found that 40 % of models had pedigrees ranging from 0.20 to 0.39, and 48 % ranged from 0.40 to 0.59. The pedigree of our Ecopath model is 0.581, and over 99.8 % of the model input parameters were effectively analyzed, which is relatively high and reflects the quality of the input parameters. Additionally, sensitivity analysis results indicated that the model's overall output values varied less than the range of variation of the input parameters. It could be due to the fact that most of our input parameters derive from referenced data. Therefore, we concluded that the Ecopath model of the KOE ecosystem can be used for strategic management considerations at the ecosystem level, similar to other ecosystem models. However, as most of the static ecosystem models, our model lacked validation with long-term data series. The data quality of the KOE Ecopath model may not be sufficient for tactical management considerations (Watari et al. 2019).

The sensitivity analysis indicated that the changes in biomass and P/B of functional groups affected all the five output indicators (TST; sum of all production, TP; calculated total net primary production, NPP; TPP/TR; total primary production/total biomass, TPP/TB). TPP is mainly affected by biomass and P/B of phytoplankton, while total biomass (TB) is affected by the biomass of each functional group, so changes in B will have an impact on TPP/TB and TPP/TR. But biomass uncertainty has a greater impact on the model's main results than P/B and Q/B, the same as other studies (Susini and Todd, 2021). Therefore, enhancing the accuracy of biomass would improve the model's accuracy. Different from most of other historical studies (Ainsworth and Walters, 2015), the field sampling information was used to compile the biomass data as a way to improve the quality of the model parameters as much as possible in this study. However, it is important to noted that our survey was conducted only once in 2023 due to constraints of ship availability and does not account for specific seasonal variations (Lam et al., 2008; Vilas et al., 2019). The representativeness of biomass data for highly migratory fish, as well as the import and export of the ecosystem, also requires careful consideration given the once-a-year survey data (Ainsworth and Walters, 2015). The trawl used in this survey consisted of a 40 mm stretch cod-end mesh with large mesh at the trawl mouth, which was not well-suited for the collection of smaller non-economic and lower trophic level species. The biomass estimates for functional groups such as anchovy, small pelagic fish, and small mesopelagic fish did not accurately reflect the actual species abundance. As a result, all of these functional groups exhibited disproportionately high EE, highlighting the need to use models to estimate their biomass more accurately. Similarly, the two functional groups of cephalopoda face a similar issue, which is primarily due to their selectivity regarding fishing gears. Therefore, improving the quality of the basic input data for functional groups such as small fishes and cephalopoda is a key focus for model optimization (Coll et al., 2013). Changes in resources of food organisms would directly or indirectly impact the food composition of predators through the food web, ultimately leading to a restructuring of fishery resources (Shapiro et al., 2022). Limited by the survey gear and schedule, we were unable to effectively estimate the abundance of the Pacific saury. The reason is that the biological characteristics of the Pacific saury, such as diurnal vertical distribution and highly migratory, make necessary a more specific sampling protocol than used here (Hua et al., 2019). Biomass and production estimates for Pacific saury were derived from the results of previous studies. Consequently, the uncertainty level of biomass of Pacific saury functional group was slightly higher. The accuracy of parameters will be further enhanced in the future through a combination of various approaches and comprehensive data analyses, such as long-term fitting of survey data and calibrations of time-dynamic simulations within the Ecosim component of Ecopath-with-Ecosim that account for seasonal variations (Christensen et al., 2005; Susini and Todd, 2021).

5. Conclusion

Based on the primary biomass data from the high-seas fishery resources survey in the KOE conducted by the research vessel Songhang of Shanghai Ocean University in 2023, we analyzed the structure and characteristics of the KOE ecosystem using the EwE model. The total system throughput (TST) and the total respiration (TR) of the KOE ecosystem were lower than those reported in most previous studies which covered more nearshore regions of the KOE. The total primary production/total respiration ratio TPP/TR was as high as 7.31, exceeding the values reported in most other studies of the KOE and its coastal ecosystems. The system omnivory index (SOI) was low (0.20), indicating that each predator group tended to feed within a relatively narrow trophic level range and the ecosystem was structurally simple and vulnerable to external disturbances. Sensitivity analysis revealed that model-derived indices were sensitive to changes and uncertainty in functional group biomass. The field sampling information was utilized as the biomass data to maximize the quality of the model. This study offers a reference for ecosystem-based fisheries management in the open ocean waters of the KOE.

CRediT authorship contribution statement

Gan Chen: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Wen Wang:** Visualization, Investigation, Data curation. **Rongrong Zhao:** Investigation. **Qiuyun Ma:** Resources, Data curation. **Shaoqin Wang:** Resources, Data curation. **Siquan Tian:** Supervision, Methodology, Conceptualization. **Dongyan Han:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the National Natural Science Foundation of China (32202934, 42306145); Program on the Survey, Monitoring and Assessment of Global Fishery Resources (Comprehensive scientific survey of fisheries resources at the high seas) sponsored by the Ministry of Agriculture and Rural Affairs, China; Open Funding Project of the Key Laboratory of Sustainable Exploitation of Oceanic Fisheries Resources, Ministry of Education.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecolmodel.2025.111152](https://doi.org/10.1016/j.ecolmodel.2025.111152).

Data availability

Data will be made available on request.

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